

# Disproof of Conjecture 00000001735

An attained maximum cannot have finitely many maximizing right-hand sides

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## 1 The statement and the counting convention

Both versions of the original conjecture assert that the maximum solution count for  $X^3 - 2Y^3 = k$  is 12 and that only finitely many  $k$  attain 12. They also promise an explicit finite list through an unspecified unit-group torsion condition. We disprove the conjunction of the maximum and finiteness assertions, which is a necessary part of the full statement.

We use the ordinary integer-solution interpretation of a Thue equation:

$$S_k = \{(x, y) \in \mathbb{Z}^2 : x^3 - 2y^3 = k\}, \quad N(k) = \#S_k, \quad k \in \mathbb{Z}.$$

Solutions are distinct ordered pairs. Signs and zero coordinates are included. Neither language restricts solutions to primitive pairs, restricts  $k$  to cube-free integers, or identifies parameters up to multiplication by cubes. None of those additional restrictions is imposed here. The customary alternative of requiring  $k \neq 0$  is also treated below.

**Theorem.** *If  $N(k) \leq 12$  for every integer  $k$  and  $N(k_0) = 12$  for some integer  $k_0$ , then infinitely many integers  $k$  have  $N(k) = 12$ . Consequently the conjecture, as stated, is false.*

*Proof.* First  $S_0 = \{(0, 0)\}$ . Indeed,  $x^3 = 2y^3$  with  $y \neq 0$  would also have  $x \neq 0$ . Taking absolute values and comparing exponents of the prime 2 in the two positive integers gives

$$3v_2(|x|) = 1 + 3v_2(|y|),$$

which is impossible modulo 3. Thus  $y = 0$  and then  $x = 0$ . In particular  $N(0) = 1$ , so a parameter  $k_0$  with  $N(k_0) = 12$  is nonzero.

For every nonzero integer  $t$ , homogeneity gives an injective map

$$S_{k_0} \longrightarrow S_{t^3 k_0}, \quad (x, y) \longmapsto (tx, ty),$$

because  $(tx)^3 - 2(ty)^3 = t^3(x^3 - 2y^3)$  and multiplication by  $t$  is injective on integers. Hence  $N(t^3 k_0) \geq 12$ . The assumed global upper bound forces  $N(t^3 k_0) = 12$ .

Take  $t = n + 1$  for  $n \in \mathbb{N}$ . The parameters  $(n + 1)^3 k_0$  are pairwise distinct: one may cancel the nonzero  $k_0$ , and cubing integers is injective. They form an infinite subset of the claimed finite set of maximizers.  $\square$

## 2 Scope of the disproof

The conclusion is the incompatibility of the two assertions. We do not determine the actual numerical maximum, prove that the number 12 is attained, or compute an actual list of maximizing parameters.

For the parameter domain  $\mathbb{Z} \setminus \{0\}$ , an upper bound of 12 extends to all integers because  $N(0) = 1$ . Excluding zero does not change the set of twelve-solution parameters. This variant is separately formalized and refuted.

The undefined torsion-condition phrase is not given an invented meaning. Whatever additional enumeration assertion it expresses, the full source still implies the maximum-and-finiteness core refuted above. In particular a finite explicit list cannot repair that contradiction.

## 3 Formalization and reproduction

The Lean project uses Lean 4.19.0 and Mathlib v4.19.0, with dependency revisions pinned in its manifest. In `lean/Solution.lean`, `Conjecture1735.solutionSet` is exactly the displayed subset of  $\mathbb{Z} \times \mathbb{Z}$  and `solutionCount` is `Cardinal.mk(solutionSetk)`. Using cardinal numbers counts the whole set even before its finiteness is known; it never assigns an artificial finite count to an infinite set. A count equal to the finite cardinal 12 is the ordinary statement that there are exactly twelve solutions.

`MaximumIsTwelve` means a universal upper bound of twelve together with actual attainment. `Conjecture` conjoins it with `Set.Finite` of the full set of attaining parameters. The main endpoints are:

- `Conjecture1735.maximum_twelve_implies_infinite_maximizers`: the asserted maximum implies `Set.Infinite` of the maximizer set;
- `Conjecture1735.conjecture_false`: the unconditional negation of `Conjecture`;
- `Conjecture1735.nonzero_conjecture_false`: the unconditional negation of the nonzero-parameter variant.

The zero-fiber argument, scaling map, injection, cardinal inequality, and infinite range are all proved in Lean. No search bound, numerical experiment, auxiliary mathematical program, or unproved finiteness theorem is used. The submitted inspection file expands the definitions and prints endpoint types and axiom dependencies. The portable verifier rebuilds the project, replays the inspection and full compiled-declaration audit with warnings treated as errors, and checks the source, dependency, and declaration records. Its accompanying controls test the rejection of invalid or altered verification inputs. Exact commands and results are recorded in `README.md` and `VERIFICATION.md`.

The original bilingual source is included as `ORIGINAL.md`, with its exact hash recorded in the verification manifest. The author derived the scaling argument from that source before an accidental exposure to unrelated task-status summaries; the access record discloses that event. A fresh reviewer independently identified the same argument before reading the proof. Full provenance and local verification records accompany the submission; local validation is distinct from maintainer acceptance.